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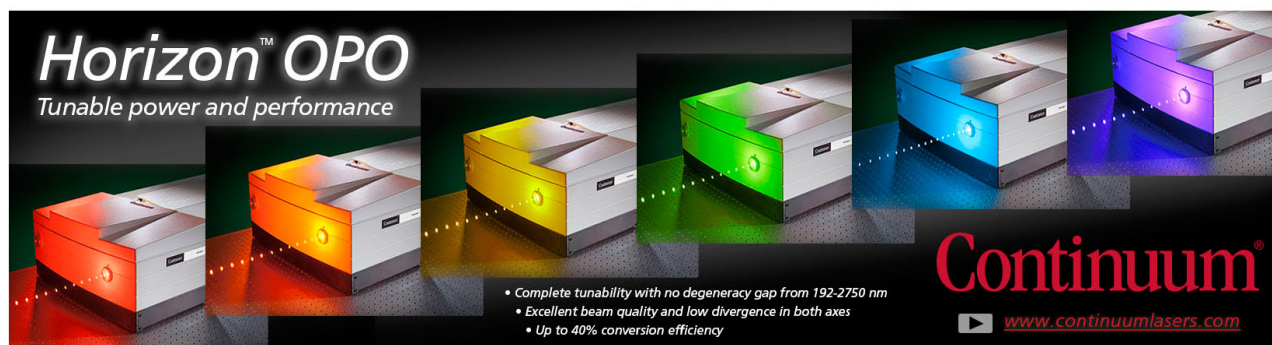
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Generalized mapping of multi-body dissipative particle dynamics onto fluid compressibility and the Flory-Huggins theory

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In this work, a generalized relation between the fluid compressibility, the Flory-Huggins interaction parameter (χ), and the simulation parameters in multi-body dissipative particle dynamics (MDPD) is established. This required revisiting the MDPD equation of state previously reported in the literature and developing general relationships between the parameters used in the MDPD model. We derive a relationship to the Flory-Huggins χ parameter for incompressible fluids similar to the work previously done in dissipative particle dynamics by Groot and Warren. The accuracy of this relationship is evaluated using phase separation in small molecules and the solubility of polymers in dilute solvent solutions via monitoring the scaling of the radius of gyration (R_g) for different solvent qualities. Finally, the dynamics of the MDPD fluid is studied with respect to the diffusion coefficient and the zero shear viscosity. © 2015 AIP Publishing LLC. [<http://dx.doi.org/10.1063/1.4919303>]

I. INTRODUCTION

The computational capabilities have advanced tremendously over the past few decades, making a significant contribution to our current understanding of the molecular scale characteristics of matter. Yet, there are clear limitations for the microscopic and atomistic level simulation techniques to capture the features of real phenomenon. For example, Molecular Dynamics (MD), one of the most popular simulation techniques, is limited to a few nanometers and milliseconds in length and time, respectively. For the same reason, methods of this scale are frequently employed to study the microstructure and physics of phenomena. On the other hand, macroscopic methods such as Finite Element Method (FEM) are unable to look at the microscopic evolution and are usually used to discuss the bulk behavior of the material subject to simulations from the continuum mechanics perspective. Thus, there is a clear need for simulation techniques that lie in between these two different scales, i.e., in order to access length and time scales larger than those accessible to microscale methods, but smaller than the lower limit in macroscale ones. These simulation techniques are usually called “mesoscale” methods. The mesoscale time and length scales are accessible by effective Coarse-Graining (CG) of the micro scales. Coarse-graining is a process in which many small particles are grouped together to form a larger bead. During the coarse-graining, small particles lose their degree of freedom to an extent which is the relevant degree of freedom for the study at hand. Perhaps, the most well-known example of this process is coarse-graining of MD particles. However, one would need to make a very careful choice of mesoscale parameters since the meaning of different parameters fundamentally change by coarse-graining. For example, instead of simulating a real material/phenomenon, one can still reproduce a behavior through parameter manipulation without a clear definition of

parameters. Thus, although CG techniques provide an effective way to extend the time and length scales, a potential limitation of these methods is lack of a clear map to actual materials and their characteristics.

Coarse-Grained Molecular Dynamics (CGMD) as an extension to traditional MD simulations provides a larger time/length scale compared to atomistic scales. Nevertheless, since CGMD in general utilized hard Lennard-Jones potentials, the coarse graining level accessible to this method remains limited. This is particularly important when time-dependent fluids, e.g., polymer melts, are subject of study. Dissipative Particle Dynamics (DPD) was first proposed as a coarse-grained molecular dynamics method to study suspensions¹ and has been improving continuously ever since. The past two decades have seen an explosion of simulation studies using DPD mainly because while it uses the same concepts as MD, it allows for time and length scales that are orders of magnitude higher than MD. In DPD, the equation of motion of a particle is written based on three different forces: the random force, which includes the thermal fluctuations in the system; the dissipative force, which acts against the motion of particle and acts as the viscosity of a fluid; and the conservative force, which gives the chemical identity of each particle and the extent of pressure between the different components of a system. The dissipative and random forces together form a thermodynamic ensemble that controls the temperature in the system and can be used to simulate non-equilibrium dynamics and incorporates Navier-Stokes hydrodynamics.² However, recent developments have shown that traditional forces in DPD are not able to capture short-range hydrodynamics and thus, there is a need for a lubrication force.³ Since the conservative force accounts for the particle-particle interactions and allows the surface tension and pressure to be calculated, it has to be included, otherwise only systems of non-interacting athermal particles can be

modeled. Also, tunable compressibility is only possible with the inclusion of the conservative force. The forces in DPD are all calculated via a weight function which basically defines the distance at which the interaction between two particles goes to zero. DPD potentials are so soft that this distance is usually defined as the separation distance between two particles (one diameter). In fact, it is this employing of very soft potentials that allow much higher scales to be achieved in DPD when compared to MD simulations.

As mentioned before, the choice of simulation parameters in DPD plays a crucial role in defining what is being represented by a particle. Since DPD parameters are dimensionless, this choice is arbitrary and not inspired by the physical characteristics of matter unless a mapping is defined and a relation between the simulation and real parameters is established. The chemical potential of a DPD particle can be correlated to the one of real matter by matching its surface tension properties. This can be done by monitoring the effect of different conservative force parameters on the interfacial tension and deriving the equation of state (EOS) based on this potential. Groot and Warren⁴ proposed a method to match different DPD terms to physical time and space parameters. They provided relations for the conservative force controlling parameter, a , of a DPD particle to be calculated based on its compressibility and the Flory-Huggins χ parameter. This has significantly extended the range of phenomena accessible by DPD simulations. In particular, by including χ , the mapping enables DPD to study dynamics of polymer systems.

In 2001, Pagonabarraga and Frenkel⁵ proposed the so-called Multi-body Dissipative Particle Dynamics (MDPD) method, based on DPD with a modified conservative force including a local density dependent term. Potentials such as a hard Lennard-Jones type inherently provide the complete van der Waals loop; however, in soft DPD interactions, the three-body interactions are mandatory for completion of a van der Waals loop, and thus the addition of the density dependent term in the conservative force makes the model capable of reproducing multiphase phenomena. MDPD has been successfully employed for various studies mainly when a multi-phase flow is of interest.^{6–11} Trofimov *et al.*¹² verified the thermodynamic consistency of this formulation and extended it to describe a strongly non-ideal, multiphase system with a correction factor. In a series of publications, Ghoufi and coworkers used MDPD to mimic various multiphase systems ranging from simple vapor–liquid mixture to polyelectrolytes and polymer brushes.^{9,10,13,14} Density dependence is necessary to describe even noble gases; however, there is no unique way to represent a pair potential that depends on density. Even though different multiphase systems can be described by the MDPD method, so far there is no report providing a robust link between the physical parameters and the MDPD terms.

For DPD, perhaps the most well-known study has been done by Groot and Warren⁴ who found a quadratic EOS in terms of the density ρ with mean-field value $\alpha = 0.101 \pm 0.001$. In case of MDPD, Merabia and Pagonabarraga^{15,16} considered one-body density dependent energies and constructed several general EOSs in terms of the structure factor

and pair correlation functions using a saddle point expansion, which predicts the introduction of a third order term for the MDPD EOS. Warren^{11,17} developed an explicit EOS for MDPD and found a third order term, but compressibility relationships and a relation to the Flory-Huggins χ parameter were not developed. This is in part due to the fact that the density dependent term in the conservative force has an additional dimension, which complicates the analysis. In a recent report, Warren¹⁸ showed that the density dependent controlling parameter, b_{ij} , has to take a constant value for different components in order for the potential to remain conservative. Other approaches have assumed an EOS or attempted to derive conservative forces from their desired EOS. Fedosov *et al.*¹⁹ assumed that the density dependence of b would add a cubic or third order term to the EOS. A more general fitting approach was adopted by Arienti *et al.*,⁷ who described the EOS as a power-law. However, in all of the above mentioned reports, a functional form of the EOS was assumed, which potentially limits the fitting capability of these studies. Although a breakthrough study on parameterization of binary mixtures via DPD was reported in Ref. 5, to date, the only report providing a bridge between the chemistry of a real fluid and MDPD parameters is the work by Ghoufi and Malfreyt,¹⁰ where a multi-scale approach is adapted based on the Flory-Huggins theory. Authors showed that by doing so, realistic representation of electrolytes with different characteristics can be obtained.

The present work proposes to address this issue by presenting a direct relationship between the simulation parameters in MDPD and experimental measures including the isothermal compressibility and the Flory-Huggins χ parameter. The development of a χ relationship for DPD enables a wide range of studies regarding material behavior, particularly, in co-polymers and polymer blends. We expect that compressibility and χ parameter relationships will allow for systematic and physically meaningful choices of conservative force parameters in MDPD and, more importantly, allow MDPD studies to use and contribute to the large literature base of studies using the Flory-Huggins χ parameter. In addition, we have studied the effect of MDPD parameters on the transport properties of the DPD fluid, namely, the viscosity and diffusion coefficient, and discussed the dependency of Schmidt number on the a and b terms. Hence, the organization of the present work is as follows: in Sec. II, a brief background on the simulation method and its formalism is given, followed by the parameters and conditions used in this study in Sec. III. Since the first step for any practical simulation-real parameter linkage to be developed is to examine the equation of state, we present the single fluid MDPD calculations and revisit the EOS in Sec. IV A. In Sec. IV B, the compressibility relationship for MDPD parameters is presented based on the results in Sec. IV A. The equations for EOS and compressibility build the foundation for the results of the binary fluids simulations, in order to obtain the Flory-Huggins interaction parameter mapping. The proposed expressions for the Flory-Huggins parameter are then validated by studying the statistical measures of a polymer chain in solvents of different qualities. Finally, we present the effect of MDPD parameters on the transport properties of a fluid.

II. BACKGROUND

A. Algorithm and simulation

DPD is governed by the Langevin equation, which is a coupled stochastic first order differential equation that describes the change in position $\mathbf{r}_i$ and velocity $\mathbf{v}_i$ of the particles in terms of $\mathbf{v}_i$ and the total forces $\mathbf{F}_i$ on the i th particle. The total force on a DPD particle is subsequently sum of three different forces and can be written as

$$\mathbf{F}_i = \sum_{j \neq i} (\mathbf{F}_{ij}^R + \mathbf{F}_{ij}^D + \mathbf{F}_{ij}^C), \quad (1)$$

where $\mathbf{F}_{ij}^R$ is the random force, $\mathbf{F}_{ij}^D$ is the dissipative force, and $\mathbf{F}_{ij}^C$ is the conservative force between particles i and j . Detailed formulation of these forces is given in Eqs. (2)–(4),

$$\mathbf{F}_{ij}^R = \sigma_{ij} \omega^R(r_{ij}) \theta_{ij} \hat{\mathbf{r}}_{ij}, \quad (2)$$

$$\mathbf{F}_{ij}^D = -\gamma_{ij} \omega^D(r_{ij}) (\hat{\mathbf{r}}_{ij} \cdot \mathbf{v}_{ij}) \hat{\mathbf{r}}_{ij}, \quad (3)$$

$$\mathbf{F}_{ij}^C = a_{ij} (1 - r_{ij}/r_c) = a_{ij} \omega^C(r_{ij}). \quad (4)$$

In Eq. (2), σ controls the strength of the thermal fluctuations and θ_{ij} is a δ -correlated Gaussian random variable. The parameter γ in Eq. (3) controls the strength of the dissipative force and acts as the viscosity for the system. Finally, the conservative force in Eq. (4) is controlled by species dependent interaction parameter, a_{ij} ,

$$\omega(r_{ij}) = \begin{cases} \left(1 - \frac{r_{ij}}{r_c}\right); & r_{ij} \leq r_c \\ 0; & r_{ij} \geq r_c \end{cases}. \quad (5)$$

All DPD potentials are calculated via a weight function (Eq. (5)) which starts at unity when two particles overlap and goes to zero at a distance between the center of two DPD particles, called “cut-off” distance. The cut-off distance in DPD is typically equal to separation distance between two particles ($= 1.0$). The weight functions in dissipative and random force are constrained by the fluctuation-dissipation relations and define the dimensionless temperature in the system as $\frac{\sigma^2}{2\gamma} = k_B T$,

$$\omega(r_{ij}) = \omega^C(r_{ij}) = \omega^R(r_{ij}), \omega^D(r_{ij}) = [\omega(r_{ij})]^2. \quad (6)$$

MDPD’s novelty and utility arise from the introduction of a density dependent attractive term to the conservative force. In this paper,

$$\mathbf{F}_{ij}^C = (a_{ij} \omega^C(r_{ij}) + b_{ij} (\bar{\rho}_i + \bar{\rho}_j) \omega^d(r_{ij})) \hat{\mathbf{r}}_{ij}, \quad (7)$$

where b_{ij} is the density dependent interaction parameter that also depends on the species involved, $\bar{\rho}_i$ is the average local density of the i th phase, and $\omega^d(r_{ij})$ is a weight function that writes the same units as $\omega(r_{ij})$ with a distance r_d for the cut-off. It should be mentioned that $a_{ij} \leq 0$ indicates the attractive nature of this term, in contrary to repulsive characteristic of $b_{ij} \geq 0$ in the conservative force. The weight function, $\omega^\rho(r_{ij})$, differs from $\omega^D(r_{ij})$ and is given by

$$\omega^\rho(r_{ij}) = \begin{cases} \frac{15}{2\pi r_d^3} \left(1 - \frac{r}{r_d}\right)^2; & r_{ij} \leq r_d \\ 0; & r_{ij} \geq r_d \end{cases}, \quad (8)$$

TABLE I. DPD-real unit conversion for r_c , m , ρ , and δt which are cut-off distance, mass, density, and time step, respectively. V is the volume of a water molecule ($30 \text{ \AA}^3$), M is the molar weight of water (18 g mol^{-1}), N_A is the Avogadro’s number, k_B is the Boltzmann constant, and T is the temperature at 298 K.

MDPD	MDPD $\rightarrow$ real units		Physical units
Bead	1	N_m	3 H_2O
r_c	1	$(\rho N_m V)^{1/3}$	6.45 ($\text{\AA}$)
M	1.00	$N_m M / N_A$	8.98×10^{-23} (g)
ρ	3.00	$\rho N_m M / N_A r_c^3$	~ 996.3 (kg m^{-3})
δt	0.01	$\delta t m^{0.5} r_c / (k_B T)^{0.5}$	~ 1 (ps)
r_D	0.75	$(\rho N_m V)^{1/3}$	4.83 ($\text{\AA}$)

where r_d is the density dependent cut-off radius, which is usually set to $0.75^{6,8,13}$ with $r_c > r_d$ to prevent discontinuities at the cut-off. Other choices of the conservative force are equally valid,²⁰ but the formulation in Eq. (8) is used for its simplicity. Also, it is important to note that the matrices of force constants for a_{ij} and b_{ij} do not change during the course of a given simulation. One can establish a parameterization scheme based on the coarse-graining level and the physical parameters, similar to one proposed by Ghoufi and Malfreyt.¹⁰ This parameterization is given in the Table I.

B. Theory and calculations

During the past decade, DPD has been used extensively to simulate polymer melts/solutions. Consequently, different models have been developed in order to study different aspects of a polymer system. For example, a so called “FENE” potential can be used in order to explain the dynamics of a polymer chain in regards to tube model.²¹ However, traditionally, this is done by the well-known bead and spring model, where groups of polymer repeat units are represented by DPD particles linked with simple spring force that serves as the covalent bond between the repeat units. The spring force usually takes the form presented in Eq. (9),

$$\mathbf{F}^S = k (r_{ij} - r_{eq}) \hat{\mathbf{r}}_{ij}. \quad (9)$$

In this, k is the spring constant and r_{eq} is the equilibrium bond length. In our simulations, we use spring constant of $k = 300$ and the equilibrium bond length of 0.65 . This is because by using the MDPD cut-off distance of 0.75 , the first neighbor peak in the radial distribution function shifts to 0.65 . Figure 1 shows the dependency of $g(r)$ on the cut-off distance.

The radius of gyration of a polymer chain as a characteristic scale reflects the effect of solvent quality on polymer size. The gyration radius of a polymer chain scales with a scaling exponent, ν , of the chain length. Flory predicted the value of 0.6 , 0.5 , and 0.33 for good, theta, and bad solvents, respectively, for the scaling exponent. The gyration radius can be calculated using Eq. (10),

$$R_g = \frac{1}{N} \sum_{i=1}^N (r_i - r_{CM})^2. \quad (10)$$

In this, N is the number of polymer beads (chain length), r_{CM} is the center of mass for the chain, and r_i is the position of i th

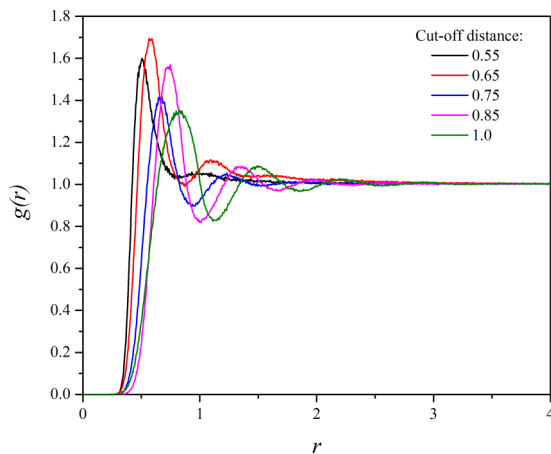


FIG. 1. Radial distribution function of DPD fluid for different MDPD cut-off values.

bead. Schlijper *et al.*²² investigated polymers in DPD using a bead-spring model in an athermal (theta) solvent and nearly obtained the correct scaling exponent for R_g of 0.50. Khani *et al.*²¹ performed the same study and found the proper scaling laws for a polymer melt. Kong *et al.*²³ considered the effect of solvent quality on R_g and found good, theta, and bad solvents by changing the conservative force parameter for the polymer-solvent interactions. Other DPD studies have been performed to reproduce scaling laws for polymer melts/solutions.^{24,25} The solvent quality and the interaction between different components of a system are best explained by the concept of free energy.

The Flory-Huggins theory is formulated in terms of the free energy per lattice site F as

$$\frac{F}{k_b T} = \frac{\varphi_A}{N_A} \ln \varphi_A + \frac{\varphi_B}{N_B} \ln \varphi_B + \chi \varphi_A \varphi_B, \quad (11)$$

where φ_A and φ_B are the volume fraction of the A and B components, respectively, and N_A and N_B are the number of segments for each A and B molecule. The free energy is minimized at

$$\chi N = \frac{\ln \left[\frac{1-\varphi_A}{\varphi_A} \right]}{1-2\varphi_A}, \quad (12)$$

which is consequently a useful way to measure χ through phase separation, when $N = N_A = N_B$. In this work, we first derive the EOS for the MDPD fluid which enables us to obtain theoretical expressions for the Flory-Huggins interaction

parameter in a similar procedure as explained by Groot and Warren.⁴ In the next step, we measure the empirical values of χ by studying the phase separation of binary fluids. To do so, we generate two different fluids of the same self-attraction and self-repulsion parameters ($a_{ii} = a_{jj}$ and $b_{ii} = b_{jj}$) with different interaction parameters ($\Delta a \neq 0$) in opposite sides of the calculation box. After reaching the steady state conditions, one can calculate the effective χ based on the density of each species far from the interface ($\phi_{A/B}$) and using the Eq. (12). We then compare the measured interaction parameters, to the ones predicted by the EOS-derived expressions.

III. SIMULATION CONDITIONS

The pressure was measured using the virial expression (with $\rho = N/V$),

$$P = \rho k_b T + \frac{1}{3V} \sum_{i=1}^{N-1} \sum_{j>i}^N \mathbf{r}_{ij} \otimes \mathbf{F}_{ij}^C$$

$$= \rho k_b T + \frac{\rho}{3N} \sum_{i=1}^{N-1} \sum_{j>i}^N \mathbf{r}_{ij} \otimes \mathbf{F}_{ij}^C. \quad (13)$$

There are a number of other ways that the pressure tensor can be measured from the interfacial tension as given by the Kirkwood-Buff²⁶ relation, the Irving-Kirkwood²⁷ method, or the test area method.²⁸ Based on the comparative study of Biscay *et al.*,²⁹ all of the methods give essentially the same pressure tensor for DPD. We chose Eq. (13) for consistency and because it gives a scalar result.

For the EOS, a simulation box with periodic boundary conditions containing 10 000 particles at densities from 3 to 10 particles per unit volume was used. The dissipative and random parameters were chosen to keep the dimensionless temperature at 1.0. The system equilibrated for 10^5 time steps of 0.01 time units before sampling. Data were then collected every time step for the next 10^6 steps. Averages with error bars for the pressure were obtained for each run. For the phase separation and computational χ measurement, a similar simulation condition was used except that the box was twice as large in the z-direction as it was in the other two directions with 6000 particles, and the phases were initialized on different sides of the box in the z-direction. This was done to reduce the interface area and extend the single phase length in the z-direction to facilitate measurement of densities far away from the interface.

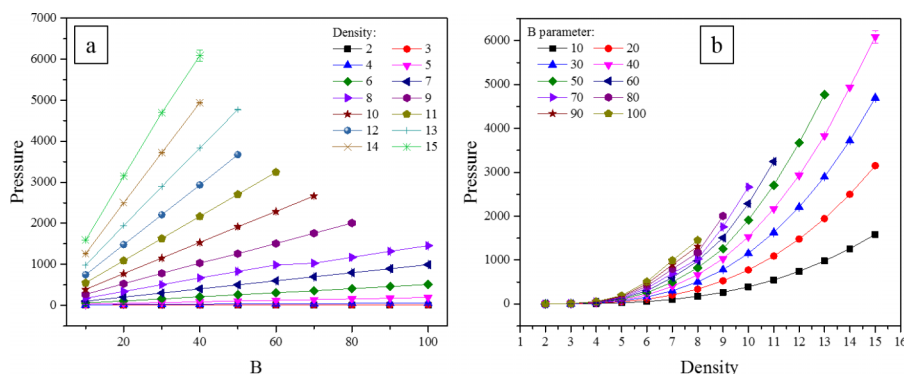


FIG. 2. Pressure as a function of (a) b_{ij} parameter and (b) particle density.

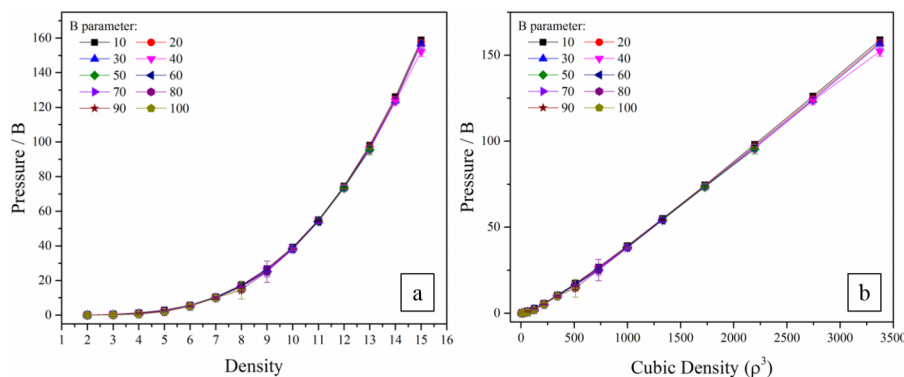


FIG. 3. Pressure divided by repulsive parameter as a function of (a) density and (b) cubic density.

For the calculation of the gyration radius, single polymer chains of varying length and 100 000 solvent particles were modeled. For these simulations, a large number of solvent particles are required for two reasons: first, the scaling laws are valid only in the dilute regime and second, the calculation box has to be large enough to avoid any polymer-polymer interaction by the imaginary neighboring cells introduced in the periodic boundary conditions.

IV. RESULTS AND DISCUSSION

A. Equation of state

In order to link the interaction parameters used in the MDPD formulation to real parameters, one would need to start from derivating their effect on the excess pressure of the system; however, this has to be done by considering the fact that having a negative a_{ij} parameter does not allow calculations consisting of only a_{ij} parameters to be performed, and thus the effect of this term has to be studied after being decoupled from the b_{ij} term (if possible, which will be discussed in detail shortly). Thus, we first performed simulations with only the repulsive (b_{ij}) term to derive proper b_{ij} -dependence of the pressure. It should be mentioned that the $b_{ij} = 0$ case, which corresponds to the standard DPD model, has been studied extensively before, and there is a second order density dependence with a_{ij} , which agrees with previous calculations. Figure 2 shows the total pressure as a function of b_{ij} and density.

The linear curves in Figure 2(a) show that in the absence of an attractive parameter, the pressure of a MDPD fluid depends directly on the value of the repulsive parameter at any specific density. This suggests that one can obtain a mastercurve by

simply dividing the measured pressure by the b_{ij} parameter. Figure 3 shows the pressure divided by the b_{ij} parameter for different densities. The results in Figure 3(b) show a cubic density dependence of the pressure, which is in agreement with previous studies.

Based on the results in Figure 3, we derive the repulsion dependence of the pressure as

$$P = \rho k_B T + 2\alpha b_{ij} r_d^4 (\rho^3 - c\rho^2 + d\rho), \quad (14)$$

where $c = 4.69$, $d = 7.55$, and $\alpha = 0.101$. Now, with the proper correlation between the pressure and the repulsive term at hand, one can seek the same relation for the attractive term of the conservative force. To do this, we have performed the simulations of varying attractive parameter with three different repulsive terms of 10, 30, and 50. The results of these simulations are plotted in Figure 4.

It should be noted that the results in Figure 4 include the calculations at which the measured pressure is a positive value. This is particularly important because as soon as a negative pressure is developed in the calculation box (which is an unphysical phenomenon), the MDPD fluid begins to retract and the effective volume of the fluid becomes smaller than the set value by the NVT ensemble. Although negative pressure values are physically possible in particular situations such as in metastable materials, in order to provide a physically meaningful representation of an experimental fluid, one can only obtain positive pressures. Thus, regardless of the possibility of a negative pressure, the measurement of this quantity does not remain valid. In order to decouple the partial pressure terms originated by a_{ij} and b_{ij} , we now plot the total pressure subtracted by the repulsive contribution given in Eq. (14).

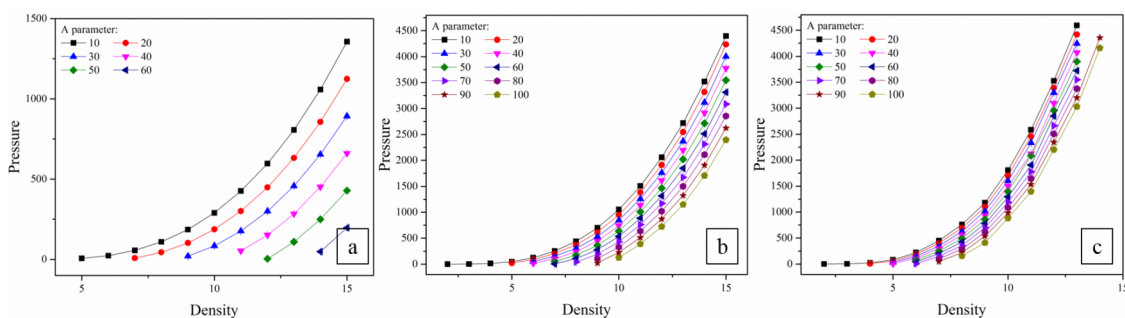


FIG. 4. Pressure vs. density for (a) $b_{ij} = 10$, (b) $b_{ij} = 30$, and (c) $b_{ij} = 50$.

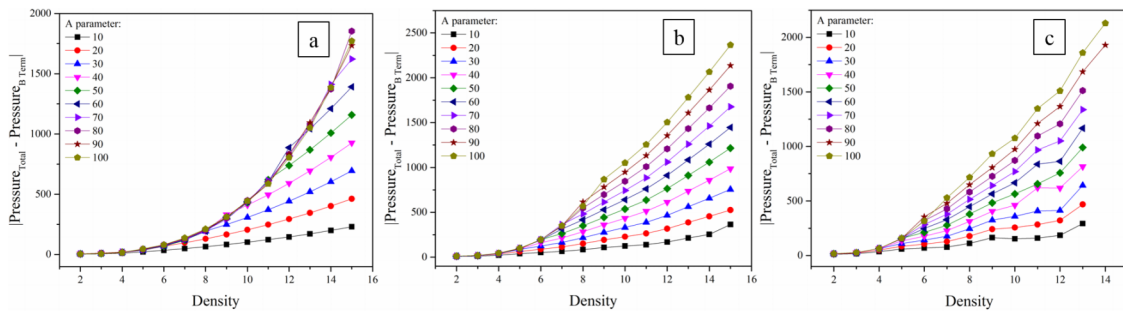


FIG. 5. Absolute value of the total pressure subtracted by the repulsive pressure as a function of density for (a) $b_{ij} = 10$, (b) $b_{ij} = 30$, and (c) $b_{ij} = 50$.

The discrepancies between the results in Figure 5 suggest that even after excluding the pressure from the repulsive term, the residual pressure still varies for different b_{ij} parameter values; however, following the same procedure as for the repulsive term, we divide the entities in the Figure 5 by the attractive parameter, a_{ij} , in order to obtain the mastercurve. The results in Figure 6 show these curves, being normalized by the extent of attractive parameter.

Equation (15) gives the expression derived from DPD and (14) as the complete equation of state for MDPD (the second term in the right hand side is the dashed line in Figure 6),

$$P = \rho k_B T + \alpha a_{ij} \rho^2 + 2\alpha b_{ij} r_d^4 (\rho^3 - c\rho^2 + d\rho). \quad (15)$$

The results in Figure 6 reveal that although the second order dependence of a_{ij} for MDPD previously reported by Warren predicts successfully predicts the pressure at low values of repulsive term, by increasing the b_{ij} parameter it underestimates the contribution of the a_{ij} term. This deviation is more pronounced at higher b_{ij} terms and is also a reverse function of the a_{ij} parameter itself. In other words, high values of b_{ij} and small a_{ij} terms lead to large discrepancies from the predicted value. This suggests that by increasing the value of B, and specifically at high densities, an a_{ij} - b_{ij} term has to be introduced in the system as well.

The discrepancies in Figure 7, in comparison with the pressure predicted by the Eq. (15), clearly show that although at high densities, namely, where the 3rd order density dependence of the repulsive term dominates the pressure, Eq. (15) successfully can describe the EOS of a MDPD fluid; however, at low and intermediate densities which are used in majority of MDPD studies, Eq. (15) overpredicts the pressure values by up to two orders of differences at the most severe

deviations. Thus, we derive the following equation which now includes the a_{ij} - b_{ij} term, in Eq. (16):

$$P = \rho k_B T + \alpha a_{ij} \rho^2 + 2\alpha b_{ij} r_d^4 (\rho^3 - c\rho^2 + d\rho) - \frac{\alpha b_{ij} r_d^4}{|a_{ij}|^{0.5}} \rho^2. \quad (16)$$

The EOS in (16) now can predict the pressure of a MDPD fluid at different a_{ij} and b_{ij} parameters, with an extra term to explain the deviations from Eq. (15). Although Warren reported 3rd order dependence of b_{ij} accompanied by 2nd order dependence of A for MDPD, his equation of state cannot properly predict the pressure in this regime. Now, with the generalized EOS at hand, the physical significance of each term in (16) can be discussed. The first term on the right-hand side of (16), $\rho k_B T$, is the term that corresponds to the ideal gas. The second term, $\alpha a_{ij} \rho^2$, originates from pairwise potentials in DPD and thus explains the two-body interactions. The third term describes the three-body interactions originating from the introduction of a density dependent term in the conservative force, which interacts with the standard two-body interactions. In fact, it is this term that allows MDPD to reflect the local densities and thus describe phase transitions as predicted. The last term on the right-hand side equation with a 2nd order dependence on density is unexpected. A previous study suggests that in density dependent conservative potentials between soft particles, this term possibly represents formation of structure or phase separation at higher densities.¹⁶ The emergence of this last term in the EOS can be explained as follows: the attractive term of the conservative force (a_{ij} term) has a weight function with the cut-off distance of 1.0, similar to traditional DPD, as opposed to the repulsive term (b_{ij}) which diminishes at the cut off distance of 0.75. Consequently, there

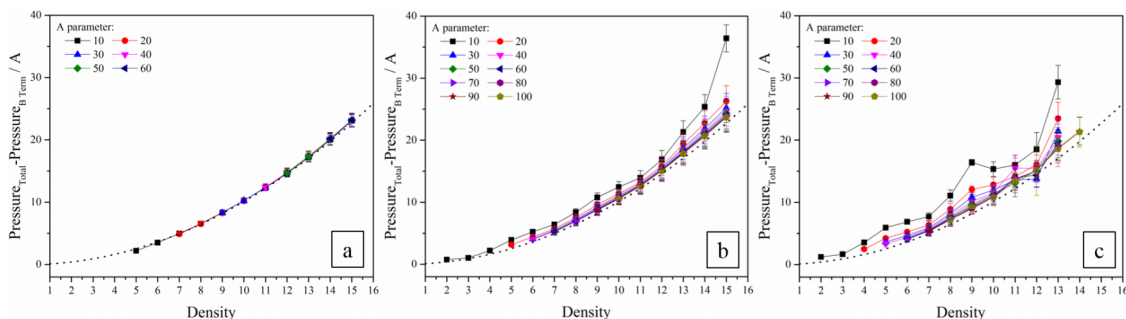


FIG. 6. Absolute value of the attractive pressure normalized by a_{ij} parameter for (a) $b_{ij} = 10$, (b) $b_{ij} = 30$, and (c) $b_{ij} = 50$. The dashed line shows the expression predicted for the conservative parameter dependence in standard DPD by Groot and Warren.

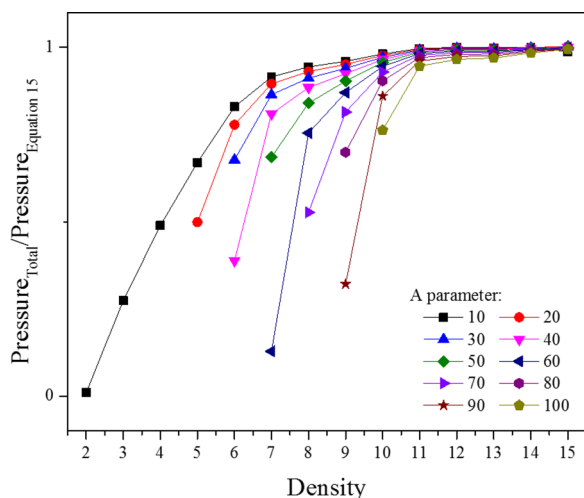


FIG. 7. Measured pressure divided by its predicted value from the Eq. (15) as a function of density.

exists a region where two neighboring particles are interacting solely via the attractive potentials. This pulls the particles to separation distances where the repulsive term becomes effective and eventually dominates the attractive term. This competition between the two opposing forces gives rise to a new term, which depends on both terms. In other words, the extent of pressure originated by the attractive term interactively changes the contribution of the pressure originated by the repulsive term as well. On the other hand, since DPD particles represent very soft spheres, the excluded volume might be significant and captured by the van der Waals. However, since this term includes a second order density dependence and is negligible at low and intermediate densities, and at high densities the pressure is completely dictated by the 3rd order term, we choose to neglect this term for compressibility and Flory-Huggins interaction parameters.

B. Compressibility and the Flory-Huggins interaction parameter

Compressibility can be used to help map the MDPD simulation parameters to the experimental fluids. As mentioned

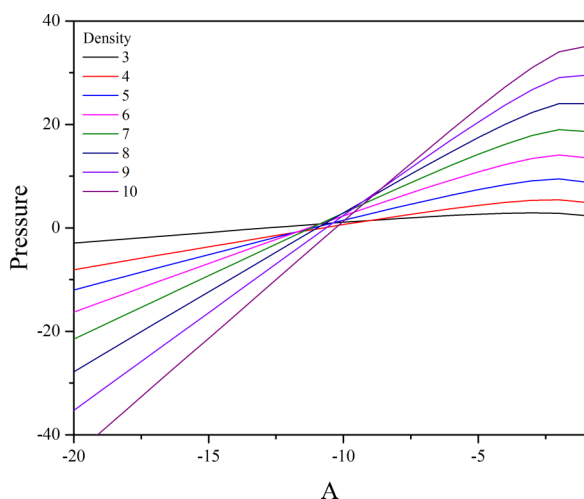


FIG. 8. Pressure as a function of a_{ij} , for a fluid with compressibility of water.

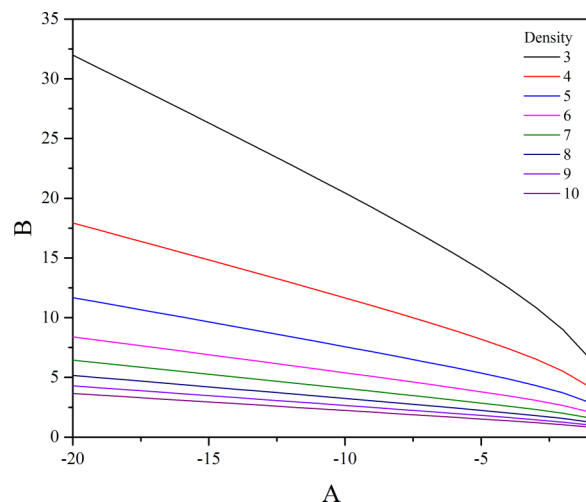


FIG. 9. b_{ij} as a function of a_{ij} based on Eqs. (16) and (17).

before, this mapping is available for DPD. For MDPD, since both a_{ij} and b_{ij} parameters contribute to the compressibility, a line is specified for compressibility, density, and temperature. The compressibility equation is calculated and presented in Eq. (17). For these calculations, the compressibility, $\kappa^{-1} = 1/(k_B T)(d\rho/dP)$, was equated to the compressibility of water ($\kappa^{-1} = 16$) and solved for either a_{ij} or b_{ij} ,

$$\frac{1}{k_B T} \frac{\partial p}{\partial \rho} = 1 + \frac{2\alpha a_{ij} \rho}{k_B T} + \frac{\alpha b_{ij} r_d^4 (6\rho^2 - 4c\rho + 2d)}{k_B T} - \frac{2\alpha b_{ij} r_d^4 \rho}{k_B T |a_{ij}|^{0.5}} \quad (17)$$

On the other hand, as mentioned before, one should carefully consider the possibility of building up a negative pressure in the calculation box. Now, we have two equations (compressibility, and also the EOS where $P = 0$) and two unknowns. Thus, based on the expression in (17), we substitute b_{ij} with an a_{ij} -dependent term in (16) and plot the pressure as a function of a_{ij} (Figure 8).

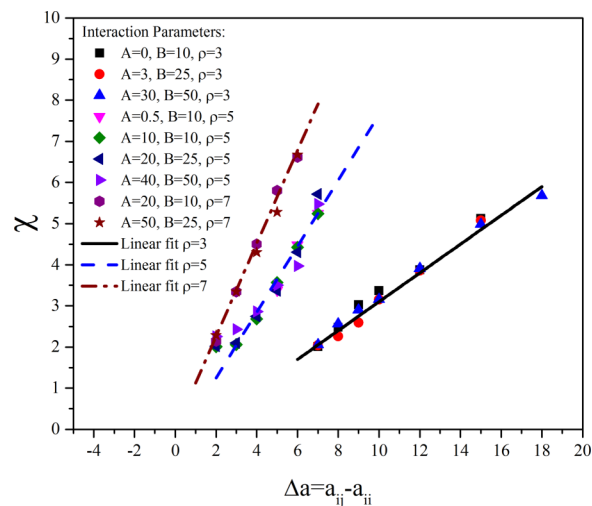


FIG. 10. Flory-Huggins interaction parameter as a function of Δa .

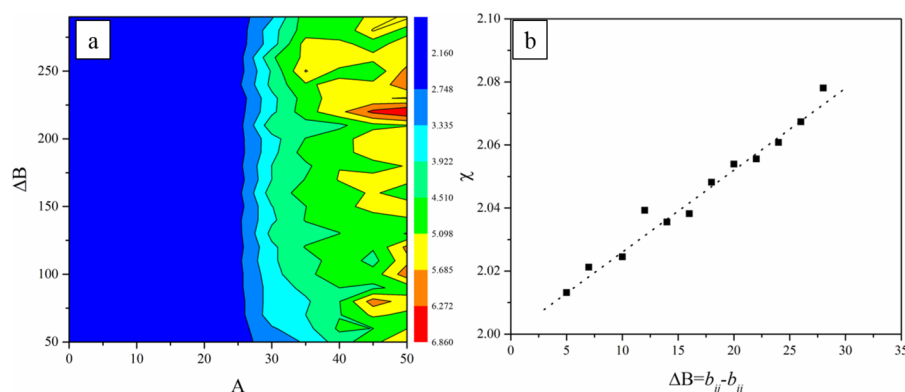


FIG. 11. (a) The color map of the interaction parameter for a range of a_{ij} and Δb choices and (b) interaction parameter as a function of Δb at constant a_{ij} parameter.

The curve in Figure 8 reveals that for a MDPD fluid to take the compressibility of water (given in (17)) and have a total positive pressure, the a_{ij} parameter cannot take values of less than 10 regardless of the fluid density. Now, we can measure the b_{ij} term as a function of a_{ij} as well (Figure 9).

Once again, it should be noted that the a_{ij} and b_{ij} pairs from the Figure 9 are the ones to first, yield a positive pressure and second, satisfy the compressibility relationship. Figure 9 also shows that in the majority of the previous studies performed via MDPD, either compressibility or pressure of the fluid was compromised as the window of accessible range to each parameter is rather narrow. Consequently, one can derive equations to calculate the Flory-Huggins interaction parameter based on a_{ij} and b_{ij} . As was mentioned before, the repulsive parameter, b_{ij} , must take a constant matrix value ($b_{AA} = b_{AB} = b_{BB}$) to remain conservative,¹⁸ and on the other hand, at low and intermediate densities, the a_{ij} - b_{ij} interaction term can be neglected. Thus, Eq. (16) can be reduced to its traditional DPD format which consequently yields the same expression for the Flory-Huggins interaction parameter, χ ,

$$\chi = \frac{2\alpha(\rho_A + \rho_B)(a_{AB} - a_{AA})}{k_B T}. \quad (18)$$

Now, in order to validate the expression for χ , one would need to first, study the phase separation of two different components at varying values of Δa and second, measure the Flory-Huggins interaction parameter as a function of Δa to confirm the validity of Eq. (18). Although a non-zero Δb is a no-go rule for MDPD, we also perform the same type of study on this parameter at a single density, to better understand the significance of the b_{ij} matrix at the most widely used simulation condition ($\rho = 3$). In practice, the phase separation phenomenon was studied by monitoring the density of each

phase at equilibrium over the calculation box and the results for Δa are shown in Figure 10. It should be noted that the mean value of x at a distance far enough from the interface was used to estimate the Flory-Huggins interaction parameter. It should be noted that since the EOS and compressibility equations provide a guideline for choosing the a_{ij} and b_{ij} parameters rather than unique solution, various pairs of parameters can be used to model the same pair of fluids. Thus at each density, several choices have been made for the particles for the interaction parameter measurements. The curve in Figure 10 shows a clear phase separation between the two fluids based on the interaction parameter.

Regardless of the choice of parameter pair choice at each density, the Flory-Huggins interaction parameter linearly increases by increasing the Δa at a constant slope. The calculated slopes are given in Eqs. (19)-(21),

$$\chi = (0.322 \pm 0.003)\Delta a; (\rho = 3), \quad (19)$$

$$\chi = (0.739 \pm 0.014)\Delta a; (\rho = 5), \quad (20)$$

$$\chi = (1.107 \pm 0.023)\Delta a; (\rho = 7). \quad (21)$$

Similar to results of Groot and Warren for DPD, the slopes measured for the interaction parameter and density deviate from the ones predicted by Eq. (18); however, this deviation for all three densities is the same ($\cong 2\alpha - 0.3$). The exponents for χ here are based on the empirical measurement of this parameter from the phase separation of binary mixtures at the interface, rather than using the EOS-derived expressions for the Flory-Huggins interaction calculations. As a result, this exponent of this parameter defers from the one suggested by Ghoufi and Malfreyt¹⁰ where the authors employed a multi-scale approach and used the FH theory to obtain this exponent.

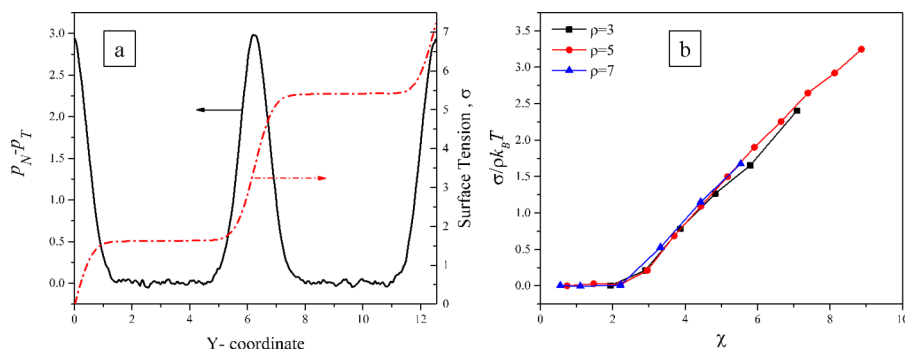


FIG. 12. (a) The difference between the normal and tangential components of the local pressure tensor along the calculation box, associated with the integral in Eq. (22) on the right axis which corresponds to the surface tension, for density of 3 and $\Delta a = 22$. And (b) normalized surface tension by the number density as a function of Flory-Huggins interaction parameter.

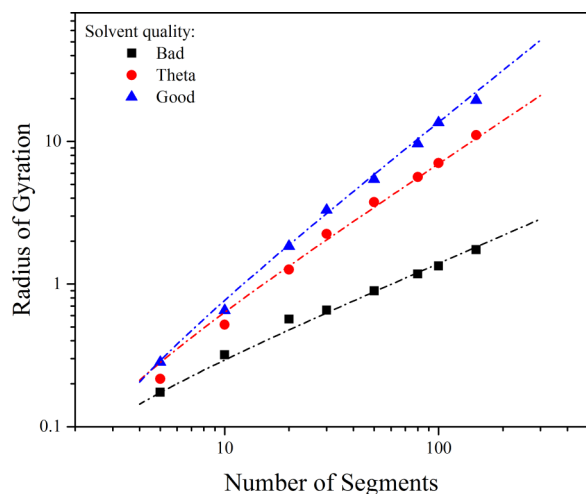


FIG. 13. Radius of gyration as a function of chain length for different solvent qualities.

Figure 11 shows the map of the interaction parameter as a function of Δb , for a range of a_{ij} choices for the density of 3.

The map of χ for the whole range of Δb and a_{ij} reveals not only that having a non-zero Δb results in a non-conservative type of potential but also for a large range it cannot effectively introduce any phase separation. The fit in the Figure 11(b) has a slope of 0.0025 which proves that changing the Δb does not significantly change the mixing behavior of the two fluids at this density. This is expected at the density of 3, because the repulsive term of the derivative of the EOS becomes negligible and thus, insensitive to the b_{ij} parameter; however, one should note that this term becomes dominant at higher densities as it includes a second order density term.

One can subsequently measure the interfacial tension of binary mixtures with different Flory-Huggins interaction parameters, in order to confirm the expressions given in Eqs. (19)–(21) and to study the stability of an interface as a function of this parameter. The surface tension can be expressed as a function of the normal ($p_N(y) = p_{yy}(y)$) and tangential ($p_T(y) = \frac{1}{2}[p_{xx}(y) + p_{zz}(y)]$) components of the local pressure tensor, and using the Irving-Kirkwood expression²⁷ as shown in the Eq. (22),

$$\sigma = \frac{1}{2} \int_{-L_y/2}^{L_y/2} [p_N(y) - p_T(y)] \cdot dy. \quad (22)$$

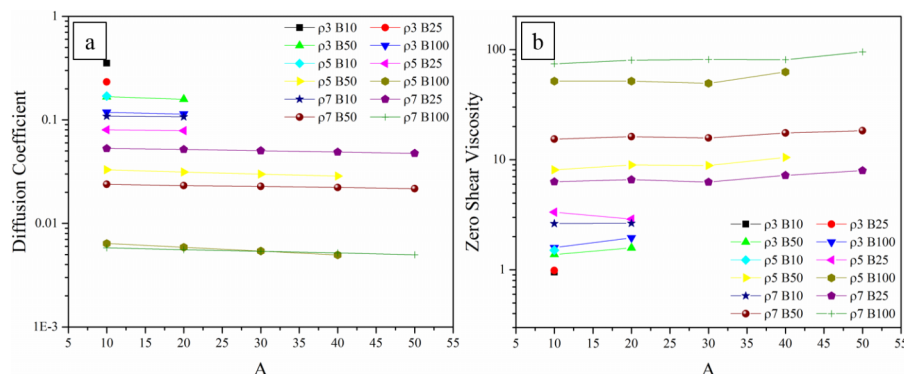


FIG. 14. (a) Diffusion coefficient and (b) zero shear viscosity of MDPD fluid as a function of repulsion parameter, a_{ij} for a range of b_{ij} parameters and densities.

Various expressions and calculation methods for the surface tension measurements in MDPD and DPD can be found elsewhere,^{9,13} where authors reported the surface tension measurements of a liquid-vapor mixture at different conditions. In our simulations, we measured the surface tension of binary mixtures with different densities and Flory-Huggins interaction parameters, in order to validate confirm the validity of Eqs. (19)–(21). Figure 12(a) shows the typical pressure components and surface tension of a binary mixture, while Figure 12(b) presents the value of surface tension as a function of χ parameter, for several densities. As theoretically predicted, the curve of the normalized surface tension given in Figure 12(b) collapses on a single curve for different densities, confirming the validity of Eqs. (19)–(21).

One limitation of phase separation measurements is that only positive Δa values can be evaluated since it is not easy to measure the difference between a perfectly mixed solution and a pair-aggregate solution from the density profiles. This motivates the use of R_g to determine the effect of both negative and positive Δa values. The scaling relationship of the R_g was investigated as a function of Δa and the results are presented in Figure 13. In these simulations, a single polymer chain with the total density of 3.0 is constructed using the bead-spring model previously discussed with $k = 300$. This is because the scaling laws for the radius of gyration are valid only in the dilute regime, where the polymer-polymer interactions can be neglected. Simulations for three Δa values over a range of chain lengths are shown in Figure 13. Since naturally a theta solvent is defined as the condition at which the polymer chain does not feel the effect of solvent, $\Delta a = 0$ is used to mimic it. Using the curves in Figure 10, $\Delta a = 10$ was used to reproduce an interaction parameter of $\chi \approx 3$ to represent a bad solvent mixture and a $\Delta a = -10$ was employed for the good solvent simulations at the density of 3. For both Δa , the expected scaling exponents are precisely captured (straight dashed lines represent theoretical values) to for the good, theta, and bad solvent with slopes of 1.18, 1.0, and 0.66, respectively.

C. Transport properties and dynamics

So far, we have studied the equilibrium and phase separation properties of MDPD fluids. Nonetheless, one of the major applications of DPD and MDPD simulations is the transport properties and the dynamics of fluids. Thus, we have performed simulations on MDPD fluids with different a_{ij} and b_{ij} parameters and calculated the zero shear viscosity and

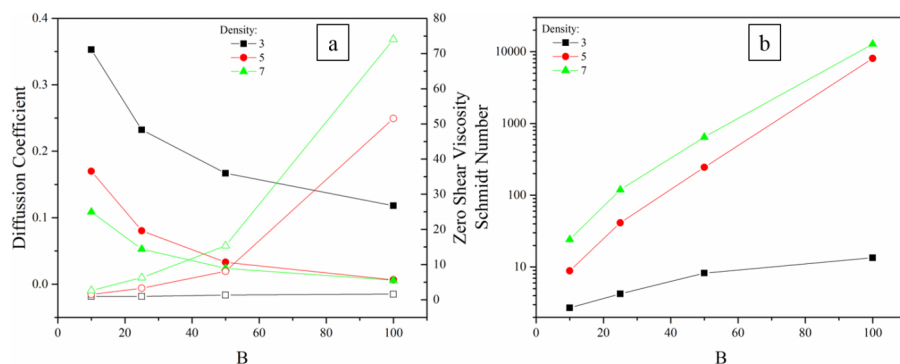


FIG. 15. (a) Diffusion coefficient (solid symbols) and zero shear viscosity (open symbols) and (b) Schmidt number of the MDPD fluid as a function of repulsive parameter, b_{ij} .

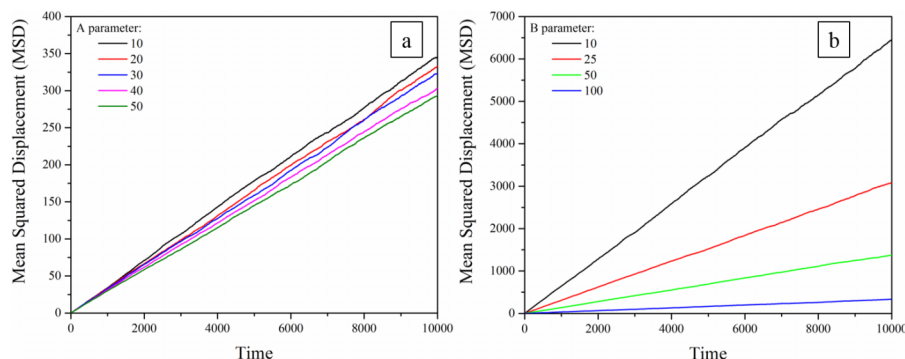


FIG. 16. Mean squared displacement of the MDPD fluid, for (a) $B = 100$ and a range of attractive terms and (b) $A = 20$ and a range of repulsive terms.

diffusion coefficient. The diffusion coefficient of a MDPD fluid can be measured using two different methods: first, by calculating the slope of the mean squared displacement of the ensemble as a function of time and second, by using the velocity autocorrelation function and following Eq. (23),

$$D = \frac{1}{3} \int_0^\infty dt \langle \mathbf{v}_i(0) \cdot \mathbf{v}_i(t) \rangle. \quad (23)$$

In the same manner, the zero shear viscosity of a fluid can be calculated from the pressure autocorrelation function using Eq. (24),

$$\eta = \frac{V}{k_B T} \int_0^\infty dt \langle \mathbf{P}(0) : \mathbf{P}(t) \rangle. \quad (24)$$

Figure 14 shows the results for the diffusion coefficient and the zero shear viscosity of MDPD fluid as a function of the a_{ij} parameter for a range of b_{ij} parameters at different densities. The results in Figure 14 reveal that over the studied range, these two main transport properties of a fluid are not a function of the attractive parameter. It should be noted that the results in these curves are limited to the simulations with positive total pressure in the calculation box.

The similar curves can be plotted for the b_{ij} parameter and subsequently using the viscosity and diffusion coefficient, one can calculate the Schmidt number as a function of b_{ij} . The results are shown in Figure 15, where a strict dependence of all transport properties on the repulsion parameter is observed. Increasing b_{ij} directly increases the viscosity of a MDPD fluid and at the same time significantly reduces its diffusion. As a result, the Schmidt number which theoretically is in the order of $\sim 10^3$ for water molecules, increases significantly

as the repulsive parameter is increased. This is particularly pronounced at elevated densities as the predominance of repulsion parameter is larger.

The insensitivity of the dynamics to the attractive parameter and at the same time, its strict dependence on the repulsive parameter can be also observed in the mean squared displacement of the MDPD fluid plotted in Figure 16. The results are from simulation at elevated densities ($\rho = 7$) and show clearly that while the Mean Squared Displacement (MSD) curves of the same b_{ij} parameter and different a_{ij} terms are within the same range, keeping the attractive term constant and changing b_{ij} significantly changes the dynamics of the system. It should be mentioned that the same type of behavior is observed for all densities.

V. CONCLUSIONS

In this paper, relations to experimental parameters have been derived for MDPD based on the Flory-Huggins interaction parameter. In order to do this, the EOS for MDPD was revisited and was found to have a third order density-dependent term for the b_{ij} parameter in agreement to prior claims; however, an additional a_{ij} - b_{ij} interaction term was found to play a crucial role at elevated densities and at low attractive parameter values. Based on this newly added term, the EOS was completed and provides a better fit at higher densities and predicts pressure within measurement errors. Based on the EOS, an expression for the compressibility of the MDPD fluid was derived and was employed in addition to the pressure measurements to define an accessible window of simulation parameters. The solution of compressibility and pressure equation at the same time for a_{ij} and b_{ij} enabled the definition of a criterion at which the compressibility of

water is satisfied and at the same time, the total pressure of the system remains positive and physically meaningful. The Flory-Huggins relationship contains a Δa term that motivates phase separation or miscibility of mixtures. The actual effect of Δa on the Flory-Huggins χ parameter was measured by monomer phase separation and R_g scaling. Although the slopes of the relationship between the Δa and χ were predicted closely by the derived expressions, the same deviations reported for DPD were observed for MDPD. Nevertheless, the derived expressions for the Flory-Huggins interaction parameter provide a very good estimate for matching the simulation parameters to the experimental characteristics of the fluid. The calculation of the gyration radius of a polymer chain, R_g , and its scaling relationships by the chain length for different solvent qualities showed that good, theta, and bad solvents could be reproduced precisely using the derived expressions.

Finally, the dynamics and transport properties of the MDPD fluid, namely, the diffusion coefficient and zero shear viscosity, were studied as a function of attractive and repulsive parameters. We found that while these entities are insensitive to the attractive term, a_{ij} , they strictly depend on the value of the repulsive parameter, b_{ij} . Thus, the dynamics of a fluid and consequently the Schmidt number can be tuned for orders of magnitude by changing this parameter. Overall, this work presents systematic choices of the MDPD conservative force parameters in order to match experimental compressibility and the Flory-Huggins χ . The χ relationship now makes the vast experimental literature using χ accessible to MDPD studies.

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